Grafana / SQLite is somewhat particular about where the database files must be located… Here is a practical guide as to how to set things up for this logger application, given Grafana and Linux conventions.

1. **Do not use your home directory for this stuff.**
2. **Make a group (I called mine 'logger') and add both youtself and grafana as group members**
3. **Everything for this cn be in the /opt/logger tree (except for working .service files**
4. **See the desired permissions after the directory structure discussion**

This is the current structure I am using

```
/opt/logger/
├── collectors/
│   ├── collect_energy.py
│   ├── collect_state.py
│   ├── state.cfg
│   ├── collect_temp.py
│   ├── temp.cfg
│   ├── pressure.txt
│   ├── .venv/
│   │   └──  #Virtual Environment-required packages: paho (MQTT), and pigpio
│   └── service_files/   #Mirrors of files that must resde in /etc/systemd/system
│       ├── collect_energy.service
│       ├── collect_state.service
│       └── collect_temp.service
└── datastores/
    ├── energy.db
    ├── states.db
    └── temperature.db
```

🔐 Permissions (simple & correct)

Keep it exactly like this:

```
/opt/logger        root:logger 750
/opt/logger/*      root:logger 750
/opt/logger/**/*.py   750
/opt/logger/**/*.txt 640

/opt/logger/datastores root:logger 770
/opt/logger/datastores/*.db 660
```

Collectors run as user in group `logger`.
Grafana is in group `logger`.

Where do the .service files go?

On a Linux system using **systemd** (including Raspberry Pi OS), `.service` files go in one of two main locations depending on scope:

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1. System-wide services (recommended for your collectors)

- **Location:**

```
/etc/systemd/system/
```

- Place your collector service files here, e.g.:

```
/etc/systemd/system/collect_energy.service
/etc/systemd/system/collect_state.service
/etc/systemd/system/collect_temp.service
```

- **Permissions:** Root-only required to create/edit.
- **Benefit:** Runs as a true system service, starts automatically on boot.

Generic Linux permission conventions -**There is no single default, but the usual convention is:**

Type	Typical Permissions
Directories	0755 (or 0700 if private)
Files	0644
Excutables	0755

🥇 Best overall: shared group ownership (recommended)

This is the clean, Linux-native way.

1. Create a shared group

```
sudo groupadd logger
```

2. Add users to it

```
sudo usermod -aG logger grafana
sudo usermod -aG logger <your-user>
```

(Log out/in after this)

Working with this structure (on the remote Rpi) from my local Linux laptop…
Use ‘sshfs’ to mount this into my local directory structure

1. **Install sshfs:**
   $sudo apt install sshfs

2. **Allow other users (such as root) to access these files:**
   $sudo nano /etc/fuse.conf

   Uncomment or add this line: user_allow_other

3. **Make a local directory (I will do this in my home directory). Establish a directory:**
   $mkdir logger
   $cd logger

4. **Connect to the remote machine:**
   $sshfs tom@logitall:/opt/logger ~/logger -o allow_other -o idmap=user

5. **To unmount these, you can do this:**
   $fusermount -u ~/logger

Setting up to run as services.

First, must make a virtual envoronment for MQTT import. I will put it here:
/opt/logger/collectors/.venv

Activate the environment:
$ source .venv/bin/activate

In this directory opt/logger/collectors/, do this:
$ sudo python3 -m venv .venv

This will be owned by root:root which we don't want. Change the permissions to allow the logger group access:
$ sudo chown -R root:logger .venv

Install the necessary import:
$ sudo /opt/logger/collectors/.venv/bin/pip install --upgrade pip
$ sudo /opt/logger/collectors/.venv/bin/pip install paho-mqtt

Now, we need .service files for each of the collectors. They go here:
`/etc/systemd/system`

Here is an example:

```
[Unit]
Description=Temperature Collector
After=network.target
Wants=network.target

[Service]
Type=simple
User=tom
Group=logger
WorkingDirectory=/opt/logger/collectors
ExecStartPre=/bin/sleep 30
ExecStart=/opt/logger/collectors/.venv/bin/python
/opt/logger/collectors/collect_temp.py
Restart=always
RestartSec=5
Environment=PYTHONUNBUFFERED=1

[Install]
WantedBy=multi-user.target
```

Just to keep thing organized, I will create these here:
/opt/logger/collectors/services

These will backup / mirror the working ones that must go in the /etc/systemd/system/ folder in order to be active.

How to use it

```
# Reload systemd so it sees the new service
sudo systemctl daemon-reload

# Enable it to start on boot
sudo systemctl enable collect_temp.service

# Start it now
sudo systemctl start collect_temp.service

# Check status / logs
sudo systemctl status collect_temp.service
journalctl -u collect_temp.service -f
```

Here are the service files in their correct place with the correct permissions:

```
 ls -l /etc/systemd/system/collect_*.service
-rw-r--r-- 1 root root 355 Jan 29 17:19 /etc/systemd/system/collect_energy.service
-rw-r--r-- 1 root root 361 Jan 29 17:19 /etc/systemd/system/collect_state.service
-rw-r--r-- 1 root root 358 Jan 29 17:19 /etc/systemd/system/collect_temp.service
```

For Reference, this is where all of my files associated with the instrumentation system are located…

Here is the logger tree:

```
$ namei -l /opt/logger
f: /opt/logger
drwxr-xr-x root root   /
drwxr-xr-x root root   opt
drwxr-x--- root logger logger
tom@logitall:/opt/logger/datastores $ tree -pug --dirsfirst /opt/logger
[drwxr-x--- root     logger  ]  /opt/logger
├── [drwxr-x--- root     logger  ]  collectors
│   ├── [drwxrwxr-x root     logger  ]  service_files
│   │   ├── [-rw-rw---- root     logger  ]  collect_energy.service
│   │   ├── [-rw-rw---- root     logger  ]  collect_state.service
│   │   └── [-rw-rw---- root     logger  ]  collect_temp.service
│   ├── [-rw-rw---- root     logger  ]  collect_energy.py
│   ├── [-rw-rw---- root     logger  ]  collect_state.py
│   ├── [-rw-rw---- root     logger  ]  collect_temp.py
│   ├── [-rw-rw---- root     logger  ]  state.cfg
│   ├── [-rw-rw---- root     logger  ]  temp.cfg
│   └── [-rw-rw---- root     logger  ]  temp.mine
└── [drwxrwxr-x root     logger  ]  datastores
    ├── [-rw-rw-r-- tom      tom     ]  energy.db
    ├── [-rw-r--r-- tom      tom     ]  states.db
    └── [-rw-rw-r-- tom      tom     ]  temperature.db

4 directories, 14 files
```

Here are the services:

```
 $ namei -l  /etc/systemd/system
f: /etc/systemd/system
drwxr-xr-x root root /
drwxr-xr-x root root etc
drwxr-xr-x root root systemd
drwxr-xr-x root root system

 $ tree -pug --dirsfirst "collect*" /etc/systemd/system

[drwxr-xr-x root     root    ]  /etc/systemd/system
├─
├── [-rw-r--r-- root     root    ]  collect_energy.service
├── [-rw-r--r-- root     root    ]  collect_state.service
├── [-rw-r--r-- root     root    ]  collect_temp.service
```